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Question	Answer	Marks
1	**Defining the problem**	
	temperature (of the water)/θ is the independent variable and volume per unit time/Y is the dependent variable **or** vary θ and determine Y	**1**
	keep temperature of room/surroundings constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • beaker open to air • water, labelled • workable method to heat water	**1**
	method to determine (change in) volume, e.g. measuring cylinder **or** method to determine (change in) mass, e.g. top-pan balance	**1**
	measure time with a stop-watch/timer (for change in volume/mass or evaporation)	**1**
	use a thermometer to measure θ **or** labelled thermometer in water in diagram	**1**
	Method of analysis	
	plot a graph of lg Y against lg θ (or ln Y against ln θ)	**1**
	s = gradient	**1**
	$k = 10^{y\text{-intercept}}$ (for ln Y against ln θ: $k = e^{y\text{-intercept}}$)	**1**

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Question	Answer	Marks
	Additional detail including safety considerations	**Max. 6**
	D1 use of (protective) gloves to handle <u>hot</u> beaker/water	
	D2 keep surface <u>area</u> <u>constant</u> (by using the same cylindrical container)	
	D3 keep water temperature <u>constant</u> (while water is evaporating)	
	D4 method to keep temperature constant while water is evaporating e.g. adjust heater/gently heat water to maintain temperature/use a water bath	
	D5 use a large surface area to increase the rate of evaporation	
	D6 $Y = \frac{\text{initial volume} - \text{final volume}}{\text{time}}$ or $\frac{\Delta V}{\text{time}}$	
	D7 $\lg Y = s \lg \theta + \lg k$	
	D8 relationship valid if a straight line	
	D9 insulation/lagging around (sides) of beaker (**not** a lid)	
	D10 switch off fans or close windows <u>to avoid draughts</u>	